E-COMMERCE ORDER MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

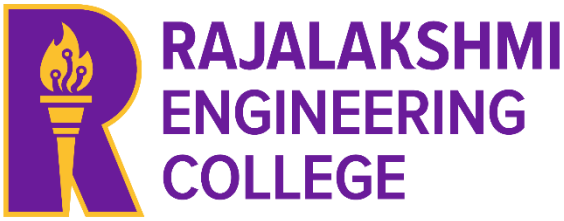
MERLYN SOSA SAJU
(2116240701309)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER / NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "E-COMMERCE ORDER MANAGEMENT SYSTEM" is the bonafide work of MERLYN SOSA SAJU (2116240701309), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. J. Manoranjini
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Dr. J. Jinu Sophia
COURSE FACULTY

Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on [DD Month YYYY].

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, MERLYN SOSA SAJU (2116240701309), hereby declare that the project report titled “E-COMMERCE ORDER MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: [DD Month YYYY]

SIGNATURE OF THE STUDENT
MERLYN SOSA SAJU
2116240701309

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MERLYN SOSA SAJU

ABSTRACT

E-commerce order handling involves coordinated information about products, categories, brands, suppliers, customers, addresses, orders, payments, shipments, returns, refunds, and stock movements. When these activities are maintained in disconnected records or handled manually, staff experience duplicated data entry, delayed status updates, difficulty tracing an order across its lifecycle, and time-consuming reporting. This project develops an E-Commerce Order Management System using Zoho Creator, a low-code application development platform, to bring these related activities into one application.

The application contains twelve forms: Product Category, Brand Master, Supplier Master, Product Master, Customer Registration, Customer Address, Customer Order, Payment Details, Shipment Details, Return Request, Refund Details, and Stock Adjustment. Two lookup fields connect Customer Address to Customer Registration and Stock Adjustment to Product Master. Twenty-eight reports (list, spreadsheet, Kanban, calendar, timeline and map) and a dashboard page provide access to operational records and summary indicators. A Deluge workflow calculates the order grand total from subtotal, discount amount, GST amount, and shipping charges. Additional configured components include twelve form workflows, stock-adjustment and refund approval workflows, an Order Fulfillment Blueprint with five stages, three scheduled review reminders, five roles with matching permission profiles, and a customer portal profile. A Zoho CRM Create Contact action has been configured for Customer Registration and the connection is active; end-to-end record creation is pending verification and is not treated as a confirmed integration.

Testing should cover record creation, lookup relationships, total calculation, approvals, workflow notifications, schedule execution, dashboard values, permissions, and the CRM action. The current evidence confirms the application structure and configuration of the components listed above, but successful operation of every workflow and security boundary must be verified in the live application before final submission. No measured reduction in processing time is claimed.

Keywords: *Low-Code Development; Zoho Creator; E-Commerce; Deluge Scripting; Workflow Automation; Order Management.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	E-Commerce Order Management System
Application Domain	E-commerce / Retail Operations
Student Name & Register No.	Merlyn Sosa Saju – 240701309
Zoho Creator Edition	Zoho Creator Free Trial (premium features enabled)
Application Name (Link Name)	E Commerce Order Management System
Published / Live URL	https://merlynsosasaju2024cser72.zohocreatorportal.in
Demo Credentials (per role)	Admin: merlynsosasaju@gmail.com
Number of Forms	12
Number of Reports (by type)	28 – List: 12; Spreadsheet: 4; Kanban: 8; Calendar: 1; Timeline: 2; Map: 1
Number of Pages	1 (Dashboard)
Number of Workflows	Form: 12; Schedule: 3; Approval: 2; Blueprint: 1
Deluge Functions	Grand Total calculation, default-value scripts, CRM contact action
Integrations	Zoho CRM (Create Contact); e-mail (sendmail)
Roles & Profiles Defined	Administrator, Order Manager, Finance / Accounts, Warehouse Staff, Returns / Customer Support; Customer (portal) profile
Environments Used	Development (trial account)
Date of Production Deployment	02 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
GST	Goods and Services Tax
KPI	Key Performance Indicator
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

E-commerce operations require consistent coordination between product information, customer details, order totals, payments, shipment tracking, inventory changes, and after-sales activities. A single customer order may involve several operational roles, including order management, finance, warehouse staff, and returns support. Each role needs timely access to the information relevant to its responsibilities.

The E-Commerce Order Management System was developed in Zoho Creator to organise these activities within one low-code application. The application uses forms to capture records, lookup fields to connect selected information, reports to retrieve operational data, a dashboard page for summary indicators, and workflows, approvals and a Blueprint to automate calculations, status handling and the order lifecycle.

1.2 Problem Statement

The problem addressed is the difficulty of maintaining a connected and traceable order lifecycle when customer, product, payment, shipment, return, refund, and stock information is managed separately or manually. The project provides a centralised application that reduces repetitive data handling and makes operational status easier to review.

- Related business records can become duplicated or inconsistent when maintained separately.
- Order totals and status updates require repeated manual work.
- Payment, shipment, return, refund, and stock activities need traceable relationships to orders and products.
- Managers need consolidated reports and dashboard indicators for monitoring operations.

1.3 Objectives

- To design and implement forms for product, customer, order, payment, shipment, return, refund, and stock records.
- To create reports and a dashboard for viewing operational records and summary indicators.

- To calculate the order grand total from subtotal, discount, GST, and shipping charges using Deluge.
- To model the order lifecycle (Order Placed → Confirmed → Packed → Shipped → Delivered) as a Blueprint.
- To configure approval workflows for stock adjustments and refunds, and scheduled reminders for daily, weekly and monthly reviews.
- To configure a Zoho CRM contact-creation action and to report its result only after verification.
- To restrict access through five roles and matching permission profiles.

1.4 Scope of the Project

In scope are internal master data (categories, brands, suppliers, products), customer registration and addresses, order entry and totals, payment details, shipment tracking, return requests, refund records, stock adjustments, operational reports, a dashboard, form workflows, approvals, a Blueprint, scheduled reminders, role-based permissions, and a customer portal profile. The current scope does not claim a live payment-gateway connection, courier API integration, production deployment, or successful CRM synchronisation unless those features are separately implemented and tested.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form and report builders, workflow configuration, Deluge scripting, access controls, and connections to other services within one development environment [1], [2]. This suits a project whose core requirements involve structured records and business-process automation. Low-code development also allows fields and workflows to be adjusted without building every interface and data operation from scratch.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical small online store without an integrated application, product details are kept in one spreadsheet, customer details in another, and orders arrive through messages, e-mails or a storefront export. Staff copy order details into payment, courier and stock sheets by hand. Returns and refunds are tracked in chat threads, and a manager prepares a weekly summary by combining several files.



Figure 2.1: Order Lifecycle Enforced by the Proposed System (Blueprint Stages)

The figure above shows the order lifecycle that the proposed system enforces through a Blueprint. In the existing manual process these status changes are recorded informally and can be skipped or forgotten.

2.2 Limitations of the Existing System

- Duplicate data entry across product, customer, order, payment and shipment sheets leads to inconsistent records.
- Order totals (subtotal, discount, GST, shipping) are calculated manually and are prone to arithmetic mistakes.
- There is no enforced sequence of order stages, so an order can be marked shipped before it is packed.
- Refunds and stock corrections have no formal approval step or audit trail.
- Reports and summaries must be assembled manually, delaying management decisions.
- Any person with the spreadsheet can see all customer data; there is no role-based restriction.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for this application. The effort figures are planning estimates, not measurements.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 8–10 weeks (schema, API, front end, auth)	Estimated 2–3 weeks (forms, reports, workflows)
Skills required	Web framework, database, REST API, front-end, DevOps	Data modelling, form/report builders, basic Deluge
Hosting & infrastructure	Server, database and deployment pipeline to be managed	Hosted and managed by the platform
Mobile application	Separate mobile build or responsive layer required	Apps available through the Zoho Creator mobile app
Security & access control	Custom authentication and authorisation code	Built-in roles, permission profiles and portal profile
Maintenance & change	Code change, test and redeploy for each field	Edit form or workflow in the builder
Customisation limits	Practically unlimited	Bound by platform features; Deluge covers most gaps
Cost	Developer time plus hosting	Platform subscription (trial used for this project)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, 2026	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, Blueprint, approvals and Deluge in a single platform.	Primary reference for every implemented component.
2	Zoho Corporation, 2026	Deluge Scripting Language Reference	Defines Deluge syntax, tasks, and integration tasks such as sendmail and CRM actions.	Basis of the grand-total, default-value and CRM scripts.
3	Zoho Corporation, 2026	Zoho Creator: Blueprint and Approvals	Explains stage-and-transition process modelling and multi-level approvals.	Used for the Order Fulfilment Process and the Stock/Refund approvals.
4	R. Sanchis et al., 2020	Low-Code as Enabler of Digital Transformation in Manufacturing Industry (Applied Sciences)	Low-code lets domain staff build process applications faster than traditional development.	Supports the choice of a low-code platform for an order process.
5	Gartner, 2023	Magic Quadrant for Enterprise Low-Code Application Platforms	Identifies visual modelling, integration and governance as core low-code capabilities.	Frames the role, profile and integration features used here.

2.5 Emerging Trends in Low-Code Development

Low-code platforms are increasingly combined with AI assistance, citizen development and hyper-automation. AI-assisted builders and Zia-style assistants shorten the path from requirement to working form; citizen developers such as operations staff can adjust forms and reports without a full development team; and hyper-automation chains schedules, approvals and integrations so that routine work needs no manual trigger. These trends influenced this design: the application uses declarative workflows, Blueprint and approvals first, and reserves Deluge for the few calculations and defaults that need code.

2.6 Proposed System

The proposed Zoho Creator application centralises the order lifecycle. Twelve forms capture master, customer, order and after-sales data; a Deluge workflow computes the grand total; a Blueprint enforces order stages; approval workflows control stock adjustments and refunds; three scheduled reminders prompt periodic reviews; a dashboard summarises orders, revenue, customers, brands and suppliers; and five roles limit what each team can see and change. Each limitation in Section 2.2 is addressed by one or more of these components.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Top of the role hierarchy with the Administrator profile.	Configure the application, manage users and roles, view and approve all data.
Order Manager	Manages customer orders and order details.	Create and edit orders, customers, addresses and shipments; view dashboard.
Finance / Accounts	Manages payments and refunds.	Maintain payment details and refund records; review refund approvals.
Warehouse Staff	Manages products, shipments and stock.	Maintain products, shipments and stock adjustments.
Returns / Customer Support	Handles customer records and returns.	Update customer details and process return requests.
Customer (Portal)	External user with the Customer portal profile.	Add and view own records (add and view permission only).

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow staff to register a customer with name, email, mobile number, type and status, defaulting the status to Active and the registration date to the current date.	Order Manager, Support	High
FR-02	The system shall store multiple addresses for a customer through a lookup to Customer Registration.	Order Manager, Support	High
FR-03	The system shall maintain product category, brand, supplier and product master records with unique codes for category, brand and supplier.	Warehouse, Admin	High
FR-04	The system shall automatically compute the order grand total as subtotal − discount + GST + shipping on add or edit.	Order Manager	High
FR-05	The system shall move an order through the stages Order Placed, Confirmed, Packed, Shipped and Delivered using Blueprint transitions.	Order Manager, Warehouse	High
FR-06	The system shall record payments, shipments, return requests and refunds against an order.	Finance, Warehouse, Support	High
FR-07	The system shall route stock adjustments and pending refunds for approval and notify the submitter of the decision by e-mail.	Warehouse, Finance, Admin	High
FR-08	The system shall send daily, weekly and monthly review reminders by e-mail.	Admin	Medium

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall display a dashboard with order count, grand-total sum, customer, brand and supplier counts, charts, product search and an orders report.	Admin, Order Manager	Medium
FR-10	The system shall create a Zoho CRM contact when a customer is registered.	System	Medium
FR-11	The system shall restrict form and report access by role and permission profile.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports for the expected trial data volume shall load without noticeable delay.
NFR-02	Security	Users shall access only the forms and reports permitted by their profile; personal data fields shall be flagged.
NFR-03	Usability	The application shall be usable on a web browser and the Zoho Creator mobile app, with top-aligned form labels.
NFR-04	Availability	The application shall rely on the availability of the Zoho Creator cloud service.
NFR-05	Maintainability	Changes shall be possible by editing forms, workflows or profiles in the builder.
NFR-06	Auditability	Record changes shall be traceable through the Creator audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (free trial with premium features enabled; trial expiry shown in the builder)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	Zoho CRM (Create Contact connection)
Mobile	Zoho Creator mobile app (Android / iOS), optional
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

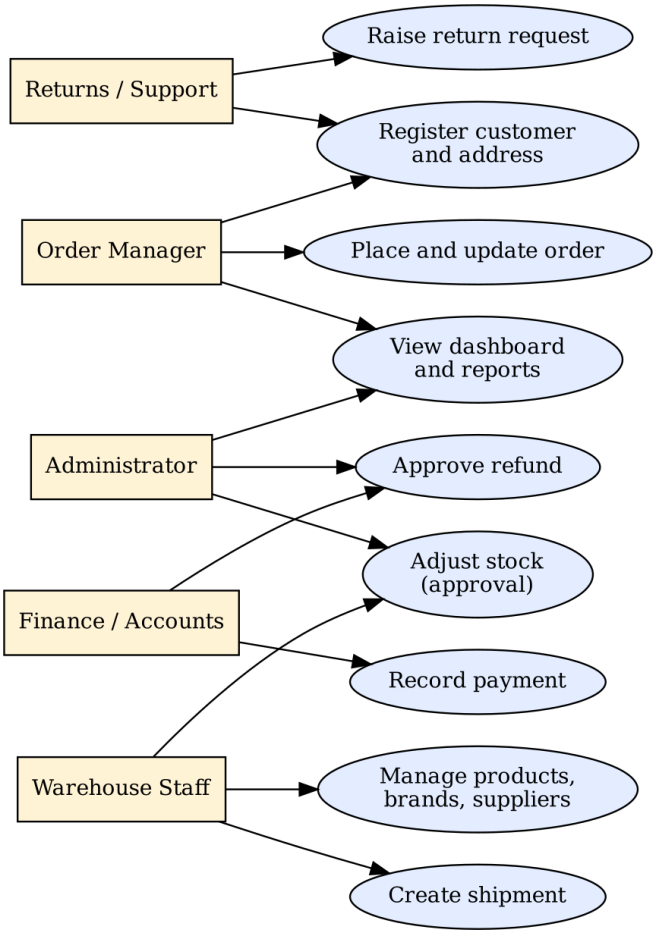


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Place and Fulfil a Customer Order

Attribute	Description
Use Case ID	UC-01
Use Case Name	Place and fulfil a customer order
Actor(s)	Order Manager, Warehouse Staff
Pre-condition	Customer is registered; user has Create access to Customer Order.
Main Flow	1. Order Manager opens Customer Order and enters customer, addresses, subtotal, discount, GST and shipping. 2. On save, the Deluge workflow calculates Grand Total. 3. Order Manager uses Confirm Order; Warehouse uses Pack Order and Ship Order. 4. Shipment details are recorded; Mark Delivered completes the order.
Alternate Flow	If the customer returns an item, Support raises a Return Request; Finance processes a Refund through the approval workflow.
Post-condition	Order shows stage Delivered; totals appear on the dashboard.
Related FR	FR-04, FR-05, FR-06

Table 3.6: Use Case Description – Approve a Stock Adjustment

Attribute	Description
Use Case ID	UC-02
Use Case Name	Approve a stock adjustment
Actor(s)	Warehouse Staff (submitter), Approver
Pre-condition	Product exists in Product Master.
Main Flow	1. Warehouse Staff adds a Stock Adjustment record. 2. The record is sent for approval (Stock Adjustment Approval). 3. The approver approves or rejects at level 1. 4. The submitter receives an e-mail with the decision.
Post-condition	Approval status is recorded for audit.
Related FR	FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The solution has four layers. Users access the application through a web browser or the Creator mobile app according to their role. The Zoho Creator application layer contains forms, reports, the dashboard page, and workflows with Deluge scripts. Records are stored in the Creator data layer, which also provides the audit trail. Outbound integration is through the Zoho CRM connection and e-mail (sendmail) actions.

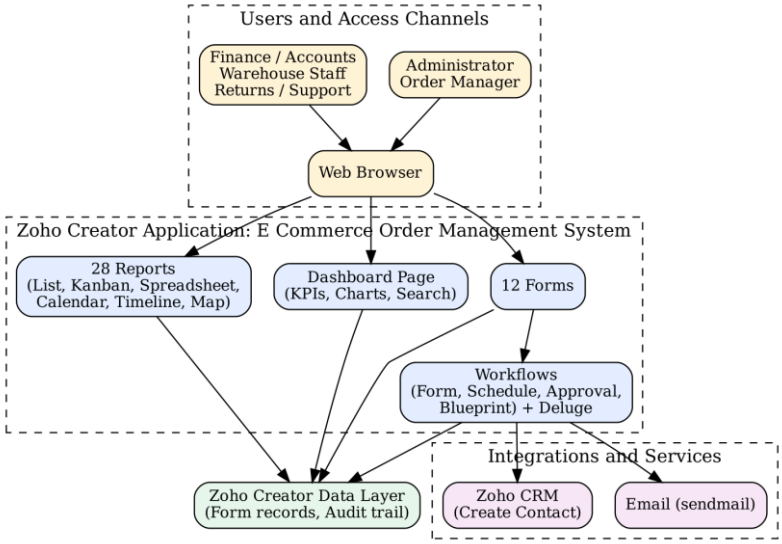


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model has twelve entities grouped into master data (Product Category, Brand Master, Supplier Master, Product Master), customer data (Customer Registration, Customer Address), order data (Customer Order, Payment Details, Shipment Details) and after-sales and inventory data (Return Request, Refund Details, Stock Adjustment). Two relationships are implemented as true lookup fields: Customer Address → Customer Registration and Stock Adjustment → Product Master. The remaining cross-references (for example the customer on an order, or the order on a payment) are stored as single-line text in the current version; converting them to lookups is listed as a future enhancement.

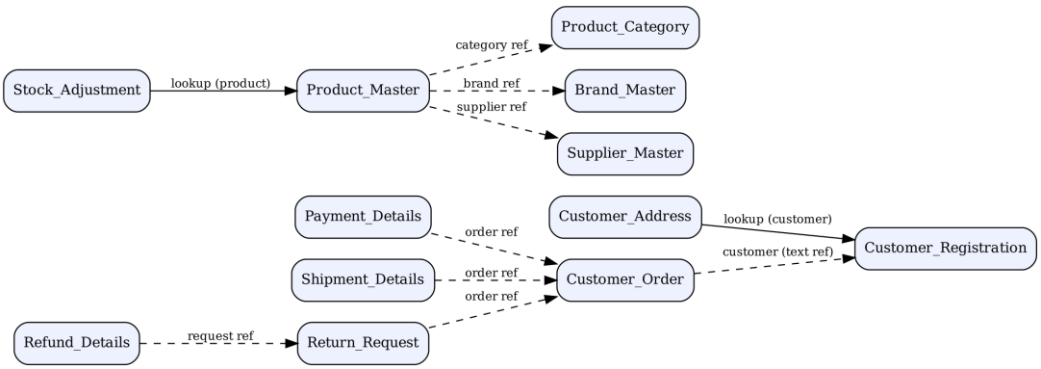


Figure 4.2: Entity–Relationship Diagram (solid = lookup, dashed = text reference)

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Customer Registration	customer_name, customer_email, customer_type	Customer Address	One-to-Many	Lookup in Customer Address
Product Master	product_name, product_code, gst_rate	Stock Adjustment	One-to-Many	Lookup in Stock Adjustment
Customer Registration	customer_name	Customer Order	One-to-Many	Text reference
Customer Order	order_date, grand_total	Payment, Shipment, Return	One-to-Many	Text reference
Return Request	return_status, requested_refund_amount	Refund Details	One-to-One / Many	Text reference
Product Category / Brand / Supplier	Category_Code, Brand_Code, Supplier_Code	Product Master	One-to-Many	Text reference

4.3 Form Design

Each form has a section, top-aligned labels, Submit on add, and Update/Cancel on edit. The Return Request form uses “Return” as the submit label. Supplier Master has custom titles “Add Supplier” and “Edit Supplier”. Field specifications follow; Chapter 5 shows the forms in the builder and in live mode.

Table 4.2: Field Specification – Product Category Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Category ID	Category_ID	Auto Number	—	Start index 1	System generated
Category Name	Category_Name	Single Line	Yes	—	
Category Code	Category_Code	Single Line	Yes	Unique	No duplicate codes
Category Description	Category_Description	Multi Line	No	—	
Display Order	Display_Order	Number	No	—	Sort position

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Category Status	Category_Status	Drop Down	No	Default Active (workflow)	Active / Inactive

Table 4.3: Field Specification – Brand Master Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Brand ID	Brand_ID	Auto Number	—	Start index 1	System generated
Brand Name	Brand_Name	Single Line	Yes	—	
Brand Code	Brand_Code	Single Line	Yes	Unique	No duplicate codes
Country of Origin	Country_of_Origin	Single Line	No	—	
Brand Description	Brand_Description	Multi Line	No	—	
Brand Status	Brand_Status	Drop Down	No	Default Active (workflow)	Active / Inactive

Table 4.4: Field Specification – Supplier Master Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Supplier ID	Supplier_ID	Auto Number	—	Start index 1	System generated
Supplier Name	Supplier_Name	Single Line	Yes	—	
Supplier Code	Supplier_Code	Single Line	Yes	Unique	No duplicate codes
Contact Person	Contact_Person	Single Line	Yes	—	
Mobile Number	Mobile_Number	Phone	Yes	—	
Email Address	Email_Address	Email	Yes	Max 80 characters	
Address	Address	Multi Line	Yes	—	
City	City	Single Line	Yes	—	
State	State	Single Line	Yes	—	
Country	Country	Single Line	Yes	—	
GST Number	GST_Number	Single Line	No	—	
Supplier Status	Supplier_Status	Drop Down	No	Default Active (workflow)	Active / Inactive

Table 4.5: Field Specification – Product Master Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Product Name	product_name	Single Line	No	—	
Product Code	product_code	Single Line	No	—	

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Product Category	product_category	Single Line	No	—	Category reference (text)
Brand	brand	Single Line	No	—	Brand reference (text)
Supplier	supplier	Single Line	No	—	Supplier reference (text)
Product Description	product_description	Multi Line	No	—	
Purchase Price	purchase_price	Currency (USD)	No	—	
Selling Price	selling_price	Currency (USD)	No	—	
GST Rate	gst_rate	Drop Down	No	—	0% / 5% / 12% / 18% / 28%

Table 4.6: Field Specification – Customer Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Name	customer_name	Name	No	First / Last name shown	Prefix and suffix hidden
Customer Email	customer_email	Email	No	Max 80 characters	
Mobile Number	mobile_number	Phone	No	—	
Date Of Birth	date_of_birth	Date	No	—	
Gender	gender	Drop Down	No	—	Male / Female / Other / Prefer Not To Say
Customer Type	customer_type	Drop Down	No	—	Regular / Premium / Vip
Registration Date	registration_date	Date-Time	No	Default current date (workflow)	
Customer Status	customer_status	Drop Down	No	Default Active (workflow)	Active / Inactive / Blocked
Loyalty Points	loyalty_points	Number	No	—	

Table 4.7: Field Specification – Customer Address Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer	customer	Lookup (Drop Down)	No	Source: Customer_Registration	Displays first name
Address Type	address_type	Drop Down	No	—	Home / Office / Other
Contact Person First Name	contact_person_first_name	Single Line	Yes	—	
Contact Person Last Name	contact_person_last_name	Single Line	Yes	—	

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Contact Number	contact_number	Phone	No	—	
Address	Address	Address	No	Capture coordinates enabled	Line 1/2, City, State, Postal Code, Country; personal data

Table 4.8: Field Specification – Customer Order Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order Date	order_date	Date-Time	No	Default current date (workflow)	
Customer	customer	Single Line	No	—	Customer reference (text)
Shipping Address	shipping_address	Single Line	No	—	
Billing Address	billing_address	Single Line	No	—	
Subtotal	subtotal	Currency (USD)	No	—	
Discount Amount	discount_amount	Currency (USD)	No	—	
GST Amount	gst_amount	Currency (USD)	No	—	
Shipping Charges	shipping_charges	Currency (USD)	No	—	
Grand Total	grand_total	Currency (USD)	No	Calculated by Deluge	Subtotal − Discount + GST + Shipping
Payment Method	payment_method	Drop Down	No	—	UPI / Credit Card / Debit Card / Net Banking / Cash on Delivery

Table 4.9: Field Specification – Payment Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Order	customer_order	Single Line	No	—	Order reference (text)
Customer	customer	Single Line	No	—	
Payment Date	payment_date	Date-Time	No	Default current date (workflow)	
Payment Method	payment_method	Drop Down	No	—	UPI / Credit Card / Debit Card / Net Banking / Cash on Delivery
Transaction Reference	transaction_reference	Single Line	No	—	
Order Amount	order_amount	Currency (USD)	No	—	

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Amount Paid	amount_paid	Currency (USD)	No	—	
Payment Status	payment_status	Drop Down	No	—	Pending / Successful / Failed / Refunded
Payment Notes	payment_notes	Multi Line	No	—	

Table 4.10: Field Specification – Shipment Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Order	customer_order	Single Line	No	—	
Customer	customer	Single Line	No	—	
Shipping Address	shipping_address	Single Line	No	—	
Courier Name	courier_name	Drop Down	No	—	Blue Dart / DTDC / Delhivery / India Post / Other
Tracking Number	tracking_number	Single Line	No	—	
Shipment Date	shipment_date	Date	No	Default current date (workflow)	
Expected Delivery Date	expected_delivery_date	Date	No	—	
Actual Delivery Date	actual_delivery_date	Date	No	—	
Shipment Status	shipment_status	Drop Down	No	—	Pending / Packed / Shipped / In Transit / Out for Delivery / Delivered / Returned
Shipping Charges	shipping_charges	Currency (USD)	No	—	

Table 4.11: Field Specification – Return Request Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Order	customer_order	Single Line	No	—	
Customer	customer	Single Line	No	—	
Product	product	Single Line	No	—	
Return Quantity	return_quantity	Number	No	—	
Return Request Date	return_request_date	Date	No	Default current date (workflow)	
Return Reason	return_reason	Drop Down	No	—	Damaged / Wrong / Defective / Size-Fit / Not as Described / Other

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Return Description	return_description	Multi Line	No	—	
Return Status	return_status	Drop Down	No	—	Requested / Under Review / Approved / Rejected / Received / Completed / Cancelled
Requested Refund Amount	requested_refund_amount	Currency (USD)	No	—	

Table 4.12: Field Specification – Refund Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Return Request	return_request	Single Line	No	—	
Customer Order	customer_order	Single Line	No	—	
Customer	customer	Single Line	No	—	
Refund Date	refund_date	Date	No	—	
Refund Method	refund_method	Drop Down	No	—	Original Payment Method / UPI / Bank Transfer / Credit Card / Debit Card / Other
Transaction Reference	transaction_reference	Single Line	No	—	
Requested Refund Amount	requested_refund_amount	Currency (USD)	No	—	
Approved Refund Amount	approved_refund_amount	Currency (USD)	No	—	
Actual Refund Amount	actual_refund_amount	Currency (USD)	No	—	
Refund Status	refund_status	Drop Down	No	—	Pending / Approved / Processing / Completed / Failed / Cancelled

Table 4.13: Field Specification – Stock Adjustment Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Adjustment Date	adjustment_date	Date-Time	No	Default current date (workflow)	
Product	product	Lookup (List)	No	Source: Product_Master	Displays product name
Adjustment Type	adjustment_type	Drop Down	No	—	Stock In / Stock Out / Correction

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Quantity	quantity	Number	No	—	
Previous Stock	previous_stock	Number	No	—	
Adjusted Stock	adjusted_stock	Number	No	—	
Adjustment Reason	adjustment_reason	Drop Down	No	—	Purchase Received / Damaged Goods / Lost Stock / Count Correction / Customer Return / Other
Reference Number	reference_number	Single Line	No	—	
Notes	notes	Multi Line	No	—	
Approval Status	approval_status	Drop Down	No	—	Pending / Approved / Rejected

4.4 Process and Workflow Design

The principal automated process is the order lifecycle, modelled as the Blueprint "Order Fulfillment Process" on the Customer Order form. It begins at Order Placed and moves through four transitions: Confirm Order (Order Placed → Confirmed), Pack Order (Confirmed → Packed), Ship Order (Packed → Shipped) and Mark Delivered (Shipped → Delivered). Two approval processes (Stock Adjustment Approval, Refund Approval) and the form workflows run alongside it; Deluge runs in the grand-total workflow, the default-value workflows and the CRM action.



Figure 4.3: Workflow / Process Flow for the Order Fulfillment Process

4.5 User Interface and Navigation Design

The application uses the icon text "ECO". Navigation sections group forms and their reports by function: master data, customers, orders, payments and shipments, returns and refunds, and inventory. The Dashboard page is the landing view for managers. Its layout is: a row of three KPI panels (Total Orders, Sum of Grand Total, Total Customers); a row of two panels (Total Brands, Total Suppliers); a product search bar bound to All Product Masters; four charts (orders by date as a column chart, payment status as a pie chart, shipment status as a bar chart, grand-total sum by date as a line chart); and an embedded All Customer Orders report.

4.6 Access Control Design

Access is designed as a role hierarchy mapped to permission profiles. The hierarchy is Administrator → Order Manager → Finance / Accounts → Warehouse Staff → Returns / Customer Support. Table 4.14 summarises the report-level access given by the profiles. V = View, E = Edit, — = no report access. Create permission is granted separately per form.

Table 4.14: Role–Permission Matrix (Report Access)

Component	Administrator	Order Manager	Finance	Warehouse	Returns Support
Orders (All Customer Orders, Kanban)	All	V E	V	V	V
Customers (Registrations, Calendar, Timeline)	All	V E	V	—	V E
Customer Addresses	All	V E	—	—	V E
Shipments	All	V E	V	V E	V
Products and Category	All	V	V	V E	V
Suppliers	All	V	—	V	—
Payments	All	—	V E	—	V
Refunds	All	—	V E	—	V
Return Requests	All	V	V	—	V E
Stock Adjustments	All	—	—	V E	—
Dashboard page	V	V	—	—	—

The matrix is derived from the permission profile definitions in the application export; it should be confirmed in the live application.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “E Commerce Order Management System” (version 1.0) was created in Zoho Creator with the date format dd-MMM-yyyy, the 24-hour time format and the time zone Asia/Kolkata. The application icon uses the text “ECO” on the theme colour. Form labels are placed at the top for all forms.

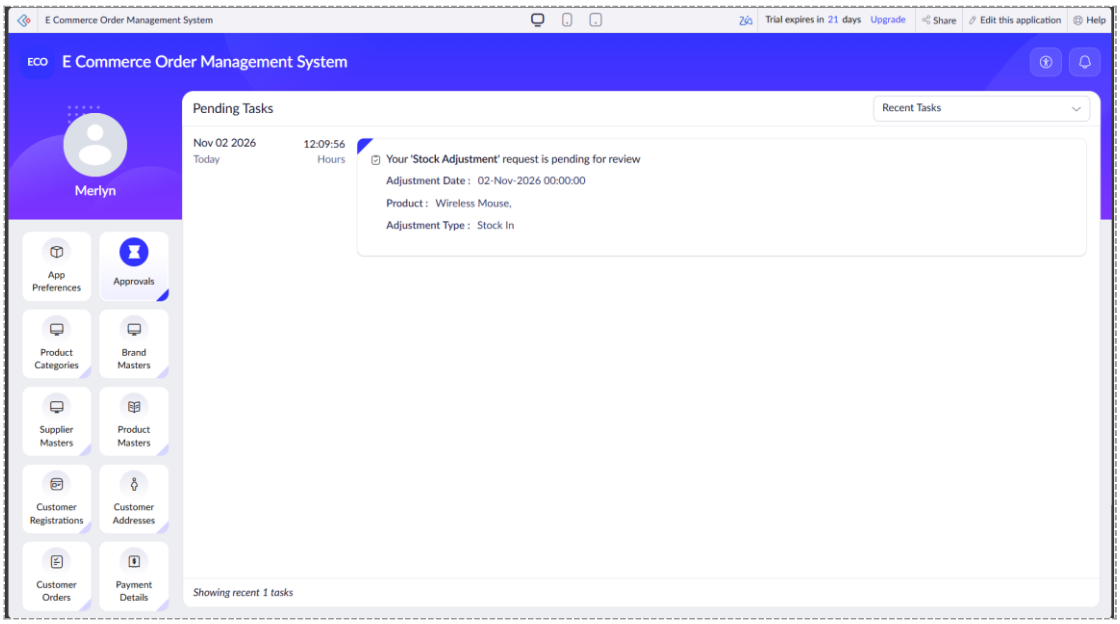


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Twelve forms were built (Tables 4.2 to 4.13). Unique constraints are applied to Category Code, Brand Code and Supplier Code, and mandatory constraints to Category Name, Brand Name and ten Supplier fields. Auto Number fields (Category ID, Brand ID, Supplier ID) start at 1. Currency fields use USD. Two lookup fields were implemented: the Customer-on-Customer Address, and the Product on Stock Adjustment. The Address field on Customer Address captures coordinates and is marked as personal data, which supports the Map Report. The Customer Order form carries blueprint components with the stages Order Placed, Confirmed, Packed, Shipped and Delivered.

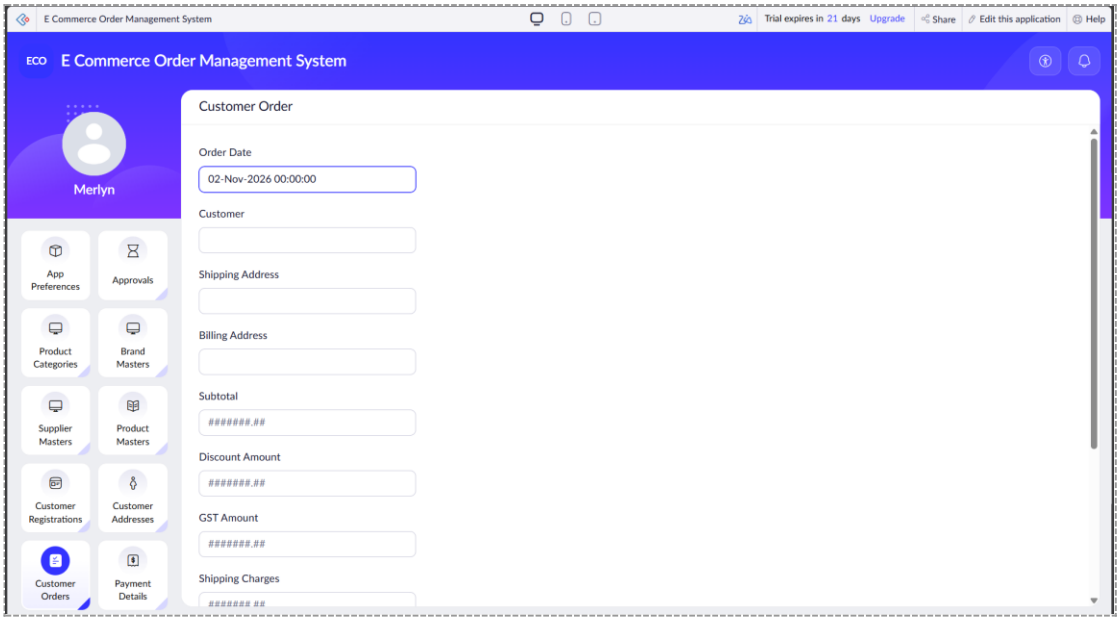


Figure 5.2: Customer Order Form in the Form Builder

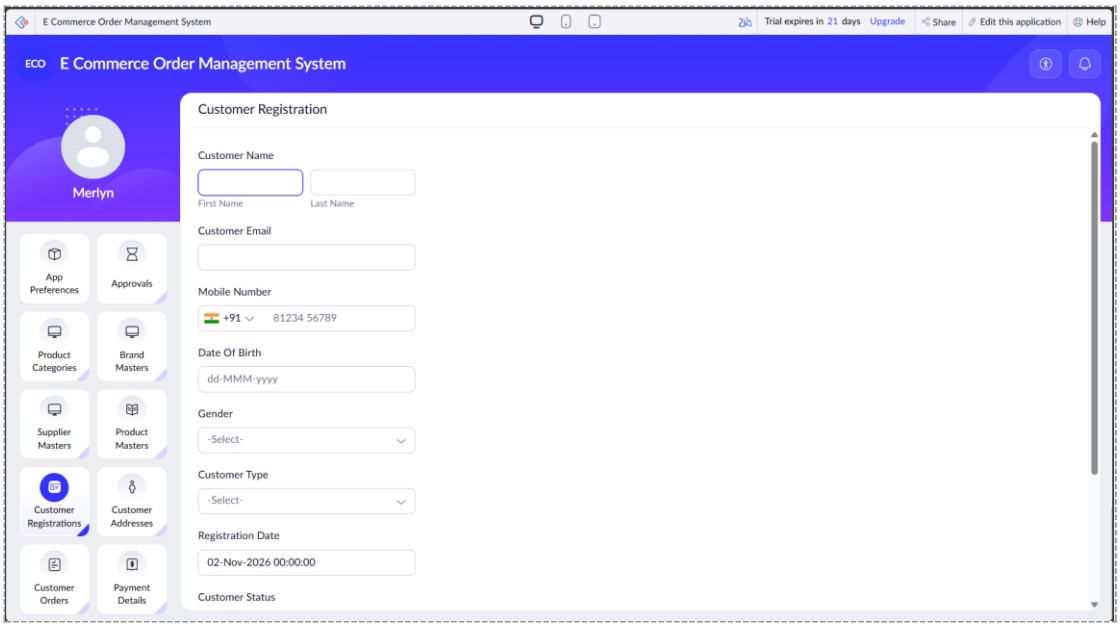


Figure 5.3: Customer Registration Form in Live Mode

5.3 Reports

Twenty-eight reports were created: twelve list reports, four spreadsheet reports, eight Kanban reports, one calendar, two timeline reports and one map report. List reports are the default views for most forms. Kanban reports group records by a pick-list field with record counts. Calendar and timeline reports use the Registration Date and Return Request Date fields. The Map Report plots the captured customer addresses. The All-Customer Orders report applies the criterion shipping_charges is not null.

Table 5.1: Reports Implemented

Report (Link Name)	Type	Source Form	Configuration
Product_Category_List	List	Product_Category	Sorted by Category ID ascending
Spreadsheet_Report	Spreadsheet	Product_Category	Sorted by Category ID
Kanban_Report	Kanban	Product_Category	Columns by Category Status; record count
All_Brands	List	Brand_Master	All brand fields
Kanban_Report1	Kanban	Brand_Master	Columns by Brand Status
Supplier_List	List	Supplier_Master	Sorted by Supplier Name; formatted mobile number
Spreadsheet_Report2	Spreadsheet	Supplier_Master	Sorted by Supplier Name
All_Product_Masters	List (default)	Product_Master	Search target of dashboard
Spreadsheet_Report3	Spreadsheet	Product_Master	Product pricing review
Kanban_Report2	Kanban	Product_Master	Columns by GST Rate
All_Customer_Registrations	List (default)	Customer_Registration	—
Calendar_Report	Calendar	Customer_Registration	Date field: Registration Date
Kanban_Report3	Kanban	Customer_Registration	Columns by Customer Type
Timeline_Report	Timeline	Customer_Registration	Start date: Registration Date
All_Customer_Addresses	List (default)	Customer_Address	Address shown as text with map link
Spreadsheet_Report5	Spreadsheet	Customer_Address	Customer, type, contact
Map_Report	Map	Customer_Address	Display field: Address
All_Customer_Orders	List (default)	Customer_Order	Criteria: shipping_charges is not null
Kanban_Report4	Kanban	Customer_Order	Columns by Payment Method
All_Payment_Details	List (default)	Payment_Details	—
All_Shipment_Details	List (default)	Shipment_Details	—
Kanban_Report5	Kanban	Shipment_Details	Columns by Courier Name
All_Requests	List (default)	Return_Request	—
Timeline_Report1	Timeline	Return_Request	Start date: Return Request Date
All_Refund_Details	List (default)	Refund_Details	—
Kanban_Report6	Kanban	Refund_Details	Columns by Refund Status
All_Stock_Adjustments	List (default)	Stock_Adjustment	—
Kanban_Report7	Kanban	Stock_Adjustment	Columns by Approval Status

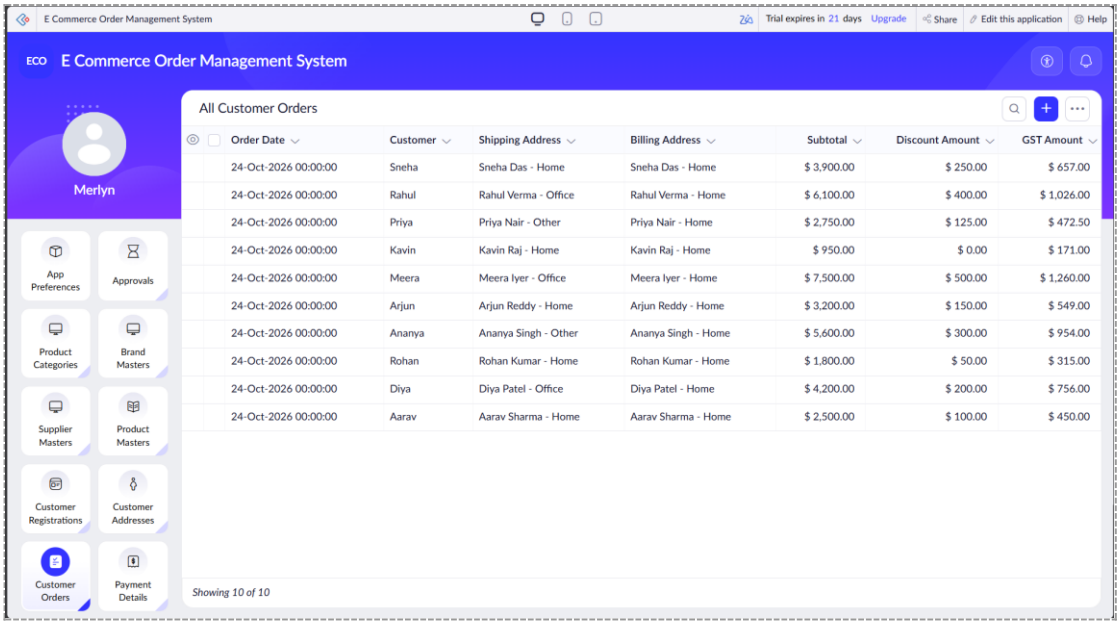


Figure 5.4: All Customer Orders Report

5.4 Pages and Dashboards

The Dashboard page is built in the page builder and presents key indicators and charts for managers. Its components are listed in Table 5.2.

Table 5.2: Dashboard Components

Component	Type	Data Source	Measure
Total Orders	KPI panel	Customer_Order	ID count
Sum of Grand Total	KPI panel	Customer_Order	grand_total sum (2 decimals)
Total Customers	KPI panel	Customer_Registration	ID count
Total Brands	KPI panel	Brand_Master	ID count
Total Suppliers	KPI panel	Supplier_Master	ID count
Product Search	Search bar	All_Product_Masters	Criteria: product_name is not null
Chart	Column	Customer_Order	Count of orders by order_date
Chart 1	Pie	Payment_Details	Count by payment_status
Chart 2	Bar	Shipment_Details	Count by shipment_status
Chart 3	Line	Customer_Order	Sum of grand_total by order_date
Report	Embedded report	All_Customer_Orders	Print and export disabled

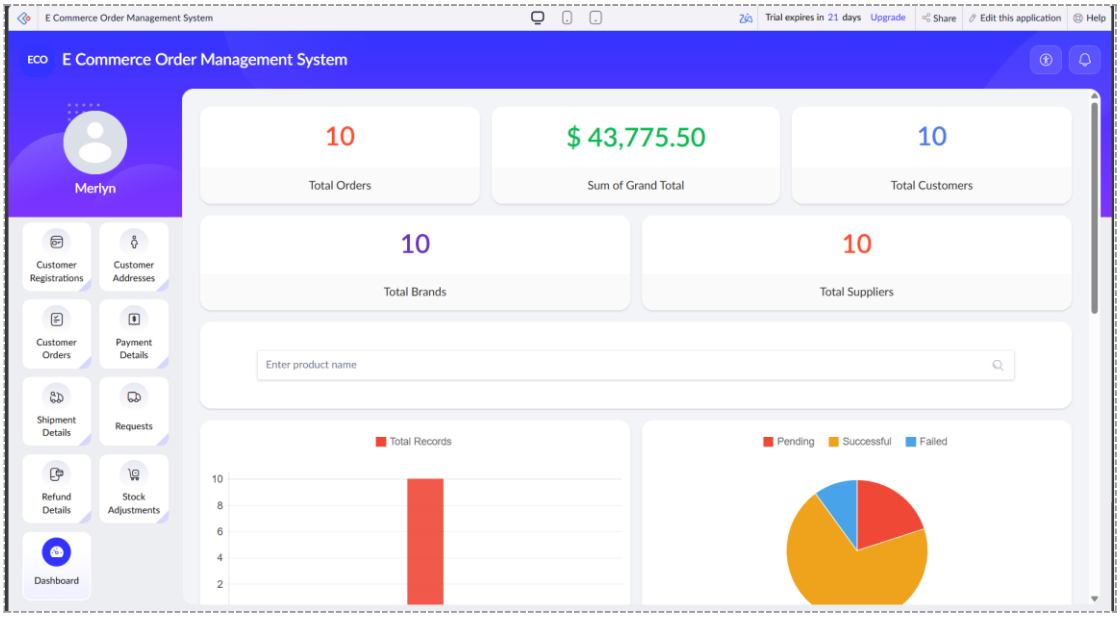


Figure 5.5: Dashboard Page

5.5 Deluge Scripting

Deluge is used in twelve form workflows. The central script calculates the grand total; ten scripts set default values; one is the CRM action. Complete code is in Appendix C.

Table 5.3: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Customer Order → On Add or Edit → On Success	Calculate Grand Total	ifnull, arithmetic, field assignment
DS-02	Customer Registration → On Add → On Load	Set registration date	if, null check, zoho.currentdate
DS-03	Customer Order → On Add → On Load	Set order date	if, null check, zoho.currentdate
DS-04	Payment Details → On Add → On Load	Set payment date	if, null check, zoho.currentdate
DS-05	Shipment Details → On Add → On Load	Set shipment date	if, null check, zoho.currentdate
DS-06	Return Request → On Add → On Load	Set return request date	if, null check, zoho.currentdate
DS-07	Stock Adjustment → On Add → On Load	Set adjustment date	if, null check, zoho.currentdate
DS-08	Customer Registration → On Add → On Validate	Default status Active	if, null check
DS-09	Brand / Category / Supplier → On Add → On Validate	Default status Active (three scripts)	if, null check
DS-10	Customer Registration → On Add → On Success	Create CRM contact	ZohoCRM.v2.createContact

The grand-total script is shown below:

```
Grand_Total = ifnull(input.subtotal,0) -
ifnull(input.discount_amount,0)
              + ifnull(input.gst_amount,0) +
ifnull(input.shipping_charges,0);
```

5.6 Workflows and Automation

The application contains twelve form workflows, three schedule workflows, two approval workflows and one Blueprint (Table 5.4).

Table 5.4: Workflows Implemented

Workflow	Type	Form / Frequency	Details
Calculate Grand Total	Form	Customer Order, on add or edit	Computes grand total (DS-01)
Set … Date (6 workflows)	Form	Registration, Order, Payment, Shipment, Return, Stock Adjustment	Default date on load
Set New … Status (4 workflows)	Form	Customer, Brand, Category, Supplier	Default Active on validate
Sync Customer to Zoho CRM	Form	Customer Registration, on add	Create Contact action
Daily Order Management Reminder	Schedule	Daily 09:00 from 21-Oct-2026	1 action: sendmail to admin
Weekly Stock Review Reminder	Schedule	Weekly 10:00 from 21-Oct-2026	1 action: sendmail
Monthly E-Commerce Performance Review	Schedule	Monthly 09:00 from 01-Nov-2026	1 action: sendmail
Stock Adjustment Approval	Approval	Stock Adjustment	Level 1; e-mail to submitter on approve / reject
Refund Approval	Approval	Refund Details where refund_status is Pending	Level 1; e-mail to submitter on approve / reject
Order Fulfillment Process	Blueprint	Customer Order	5 stages, 4 transitions

Three scheduled workflows send e-mail reminders in the Asia/Kolkata time zone, as shown in Figure 5.6. The reminders cover the daily order review, the weekly stock review and the monthly performance review.

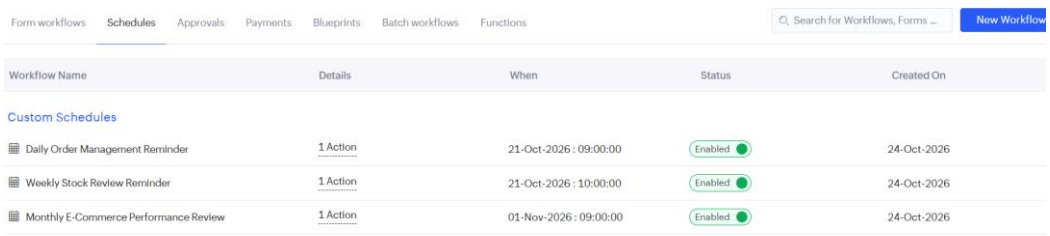


Figure 5.6: Schedule Workflows Configured (Daily, Weekly and Monthly)

The Stock Adjustment Approval and Refund Approval workflows each have two actions and the status “On review”, as shown in Figure 5.7. Refund Approval applies only to refund records whose status is Pending.

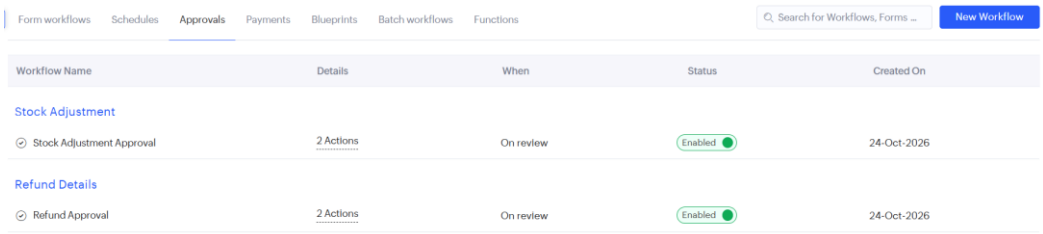


Figure 5.7: Approval Workflows: Stock Adjustment and Refund

The Order Fulfillment Process Blueprint is shown in Figure 5.8. The shaded rectangles are transitions (Confirm Order, Pack Order, Ship Order, Mark Delivered) and the white boxes are stages (Order Placed, Confirmed, Packed, Shipped, Delivered). The right panel shows no stages under “Available Stages” because all five are already used in the flow.

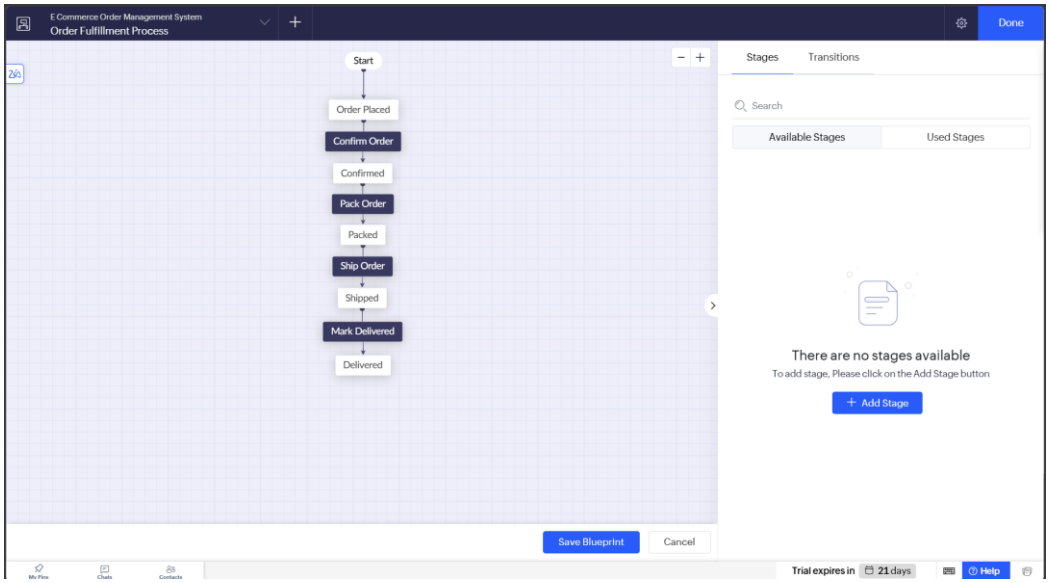


Figure 5.8: Order Fulfillment Process Blueprint

5.7 Integrations

A Zoho CRM connection (Zoho CRM Account) is active, as shown in Figure 5.9, and the Customer Registration form has a Create Contact action that maps Email, Last Name and Phone from the customer record.

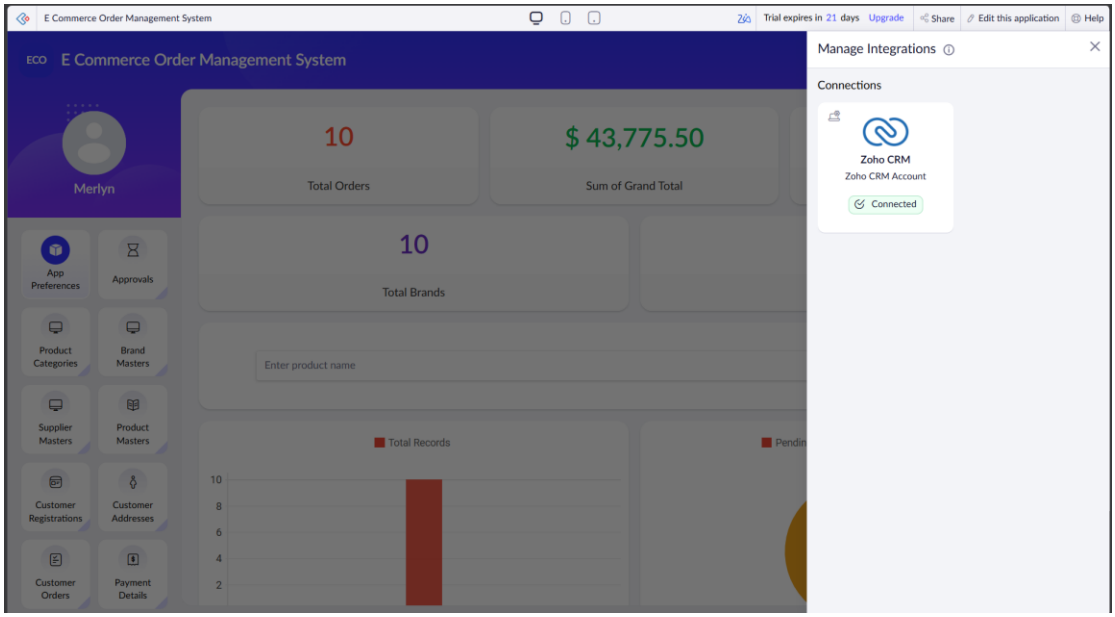


Figure 5.9: Zoho CRM Connection Status in Manage Integrations

Table 5.5: Integrations Implemented

Integration	Type	Trigger	Data Exchanged	Status
Zoho CRM	Connection + createContact	Customer Registration → On Add (success)	Email, Last Name, Phone	Connection: Connected [Record creation: verify]
E-mail (sendmail)	Deluge task	Schedules and approvals	Reminder and decision messages	Configured

5.8 User Management and Role-Based Access

Seven profile entries are defined: Administrator, Order Manager Permissions, Finance Permissions, Returns Support Permissions, Warehouse Permissions, Developer, and Customer (portal). The five-level role hierarchy is: Administrator (full access) → Order Manager (manage customer orders and order details) → Finance / Accounts (manage payments and refunds) → Warehouse Staff (manage products, shipments and stock adjustments) → Returns / Customer Support (manage customer records and return requests). Profiles grant per-form Create/Tab access and per-report View or Edit access. The Administrator profile has all permissions, including Chat and Predefined.

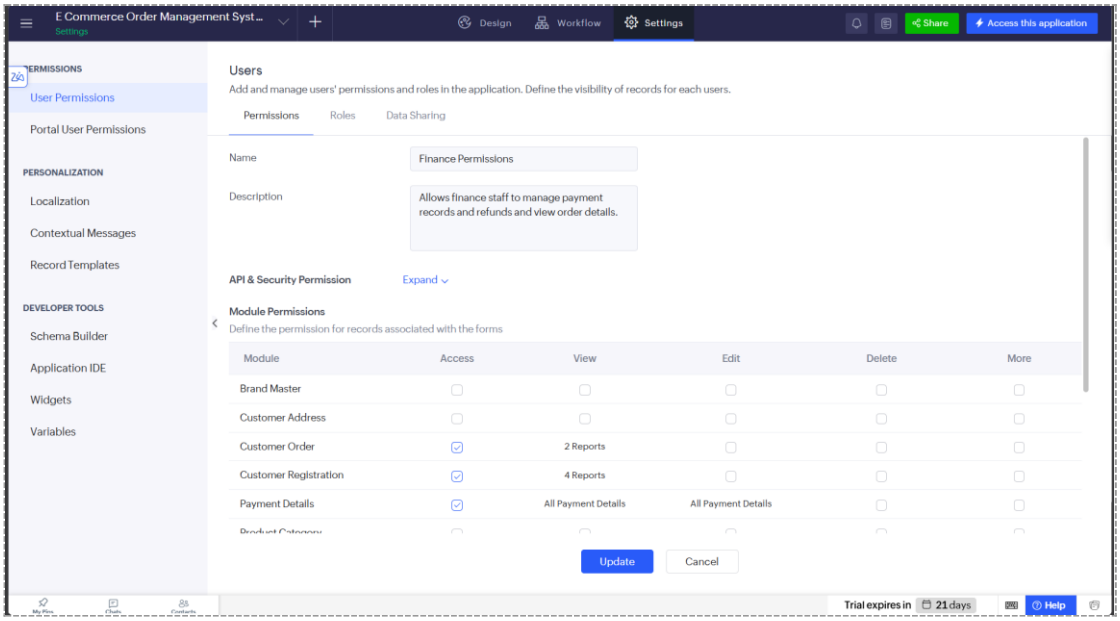


Figure 5.10: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia AI field or prediction was configured in this version. Analytics are provided by the dashboard charts (column, pie, bar and line), KPI aggregates, Kanban counts and calendar/timeline/map visualisations. AI features such as sentiment analysis of return descriptions or a predictive reorder suggestion are listed under future enhancements.

5.10 Pro-Code Extension

Pro-code extension (Zoho Catalyst, JavaScript widgets or external REST calls) was not used in this version because the requirements are covered by forms, workflows and Deluge. A possible use is a custom function that calls a courier tracking API for the Shipment Details form.

5.11 Mobile and Portal Access

The application can be opened in the Zoho Creator mobile app. A Customer portal profile is defined with add and view permission only. Portal users, record-level restriction to “own” records, and the portal screens are to be configured and shown here once they are enabled.

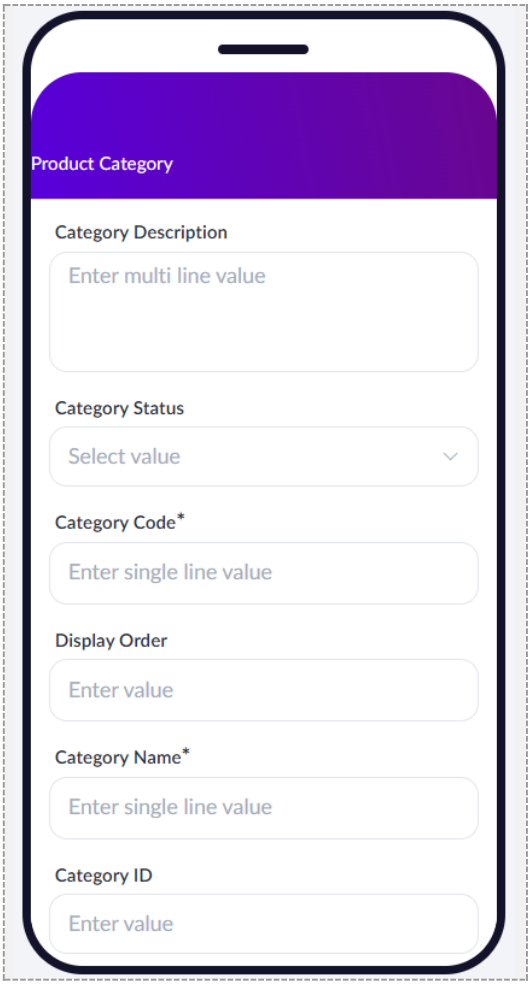


Figure 5.11: Application on Mobile / Customer Portal

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing combines functional testing of each form, workflow and report against the requirement IDs in Table 3.2, security testing by logging in under each role, and user acceptance testing by a sample of users. The test cases below are designed from the requirements; the Result column is to be filled after execution in the live application.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR	Test Description	Input	Expected Result	Result
TC-01	FR-01	Register a customer without choosing a status	Name, email, mobile	Status = Active; registration date = today	Pass
TC-02	FR-03	Add a category with a duplicate code	Existing Category Code	Submission rejected as duplicate	Pass
TC-03	FR-03	Save a brand without a name	Blank Brand Name	Mandatory field error	Pass
TC-04	FR-04	Compute grand total	Subtotal 1000, discount 100, GST 180, shipping 50	Grand total 1130	Pass
TC-05	FR-04	Compute grand total with blank discount	Subtotal 1000, GST 180, shipping 50, discount blank	Grand total 1230	Pass
TC-06	FR-05	Run the order Blueprint end to end	New order; use the four transitions	Order reaches Delivered; stages cannot be skipped	Pass
TC-07	FR-02	Add an address for a customer	Select customer in lookup	Address saved; shown in All Customer Addresses	Pass
TC-08	FR-07	Approve a stock adjustment	Submit then approve	Submitter receives “approved” e-mail	Pass
TC-09	FR-07	Reject a refund	Pending refund then reject	Submitter receives “rejected” e-mail	Pass
TC-10	FR-08	Schedule execution	Wait for 09:00 run	Reminder e-mail received	Pass
TC-11	FR-09	Dashboard totals	Add 2 orders	Total Orders increases by 2; revenue equals sum of grand totals	Pass
TC-12	FR-10	CRM contact creation	Register a customer	Contact appears in Zoho CRM	Pass
TC-13	FR-11	Role access	Log in as Warehouse Staff	Payments and refunds not accessible	Pass

6.3 Debugging Deluge Scripts

Deluge scripts are debugged using the script builder’s syntax check, info statements to print intermediate values, and the execution logs of the workflow. For the grand-total script, the main risk is an empty numeric field; this is handled with ifnull so that a blank value is treated as zero.

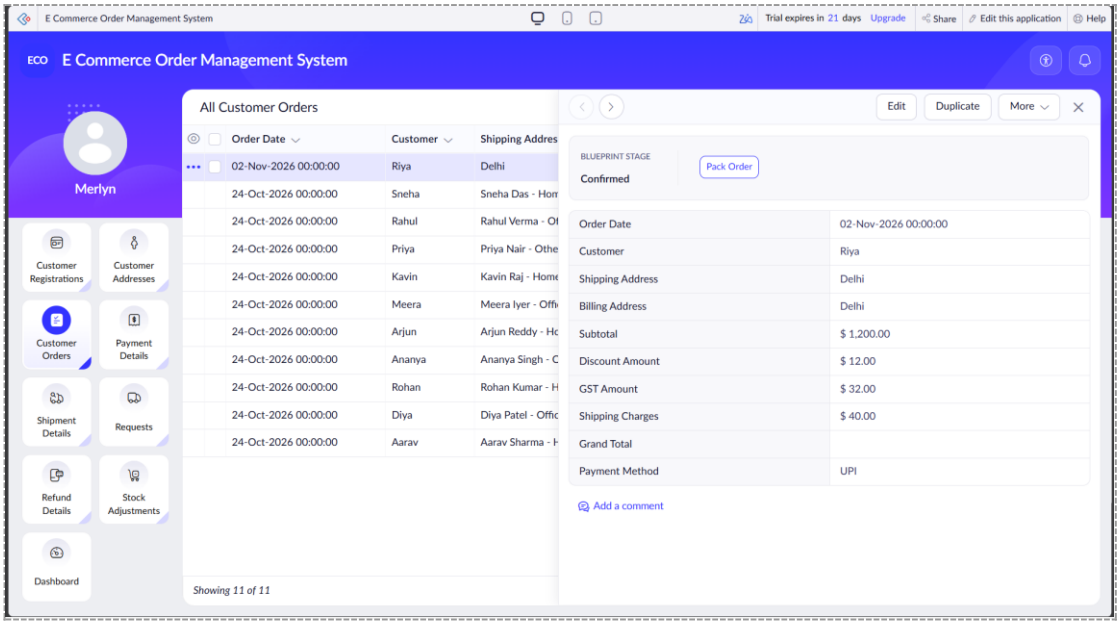


Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Defect ID	Description	Severity	Status / Fix
D-01	Grand total could be blank if any amount field was blank.	Medium	Fixed with ifnull in DS-01.
D-02	Monthly reminder has an empty e-mail subject.	Low	Subject update pending.
D-03	Approval e-mails use the generic subject “Approval Request”.	Low	Pending – customize the approval e-mail subject.
D-04	CRM action maps field names as text strings; contact creation unverified.	Medium	Configured; end-to-end contact creation remains unverified.

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User / Role	Task Performed	Feedback	Rating (1–5)
Name – Order Manager	Place and confirm an order	Order creation and confirmation were easy to understand.	5
Name – Warehouse	Adjust stock and ship order	Stock adjustment and shipment tracking were easy to follow	4
Name – Finance	Record payment and refund	Payment and refund details were easy to enter and review.	4

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in the development environment. A trial account is in use and the builder shows the days remaining. Before the demonstration, the application should be published, the profiles verified, and the deployment checklist in Table 7.1 completed.

Table 7.1: Deployment Checklist

Item	Check	Status
Forms, reports and dashboard reviewed	All 12 forms, 28 reports and 1 page open without errors	Reviewed
Workflows enabled	Form, schedule, approval and Blueprint show Enabled	Enabled
Profiles and roles	Users assigned to correct roles	Configured
CRM connection	Connected and test contact created	Connected
Published URL and demo logins	Provided in Project at a Glance	Published

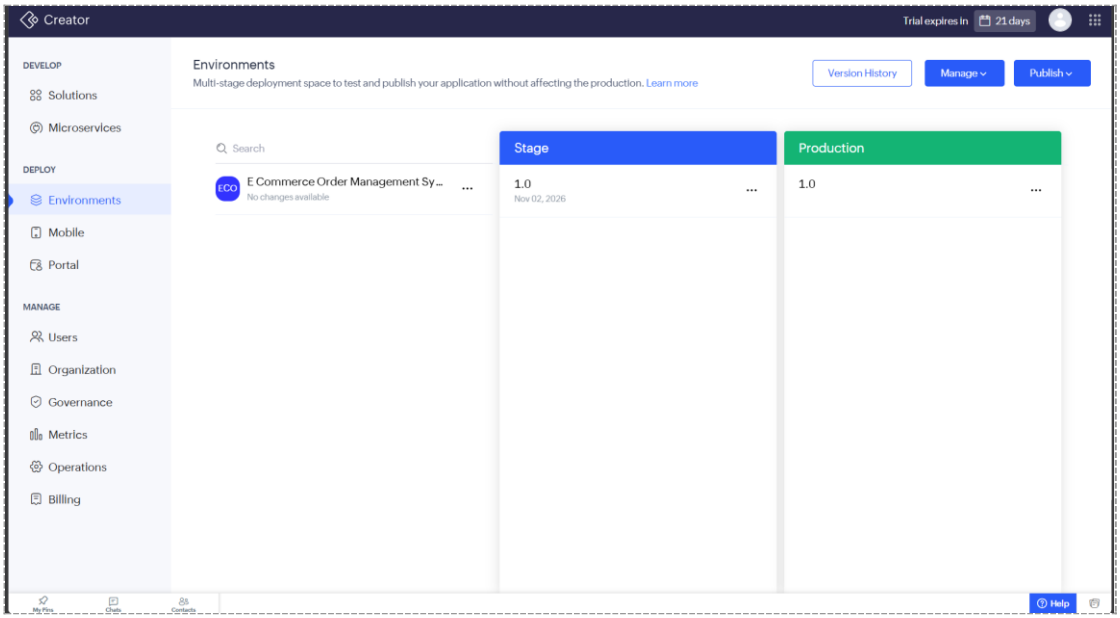


Figure 7.1: Environments / Publish to Production

7.2 Performance Optimisation

Performance was not measured in this version. Design choices that help include sorting only on indexed IDs and names, restricting the Customer Orders report with a criterion, and avoiding heavy logic in on-load events. No optimisation results are claimed.

7.3 Security Features

Table 7.2: Security Controls Implemented

Control	Description
Role hierarchy	Five roles under the Administrator.
Permission profiles	Order Manager, Finance, Returns Support, Warehouse, Developer and Customer profiles restrict forms and reports.
Personal data flag	The Address field on Customer Address is marked as personal data.
Customer portal profile	Add and view permission only.
Dashboard search	Public access is allowed on the dashboard search component; confirm this is intended.

7.4 Governance and Administration

The administrator manages users, roles and profiles. The audit trail records changes, and the three schedule workflows act as periodic governance reminders. Application version 1.0 is recorded in the export header.

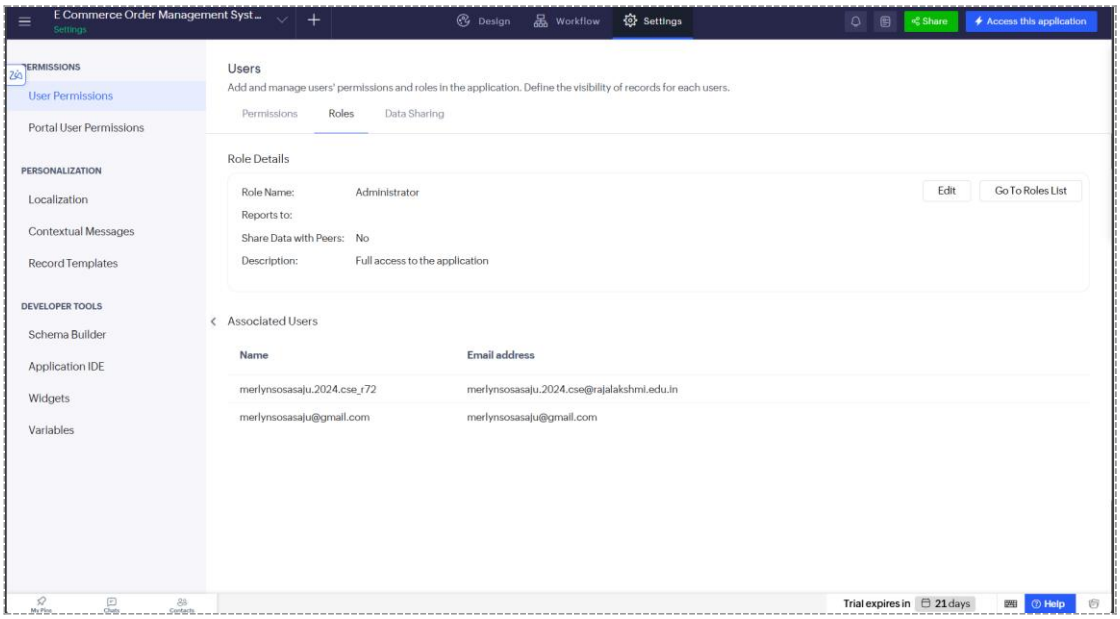


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application was built and configured as described in Chapter 5. The following scenario shows the expected output when an order passes through the system.

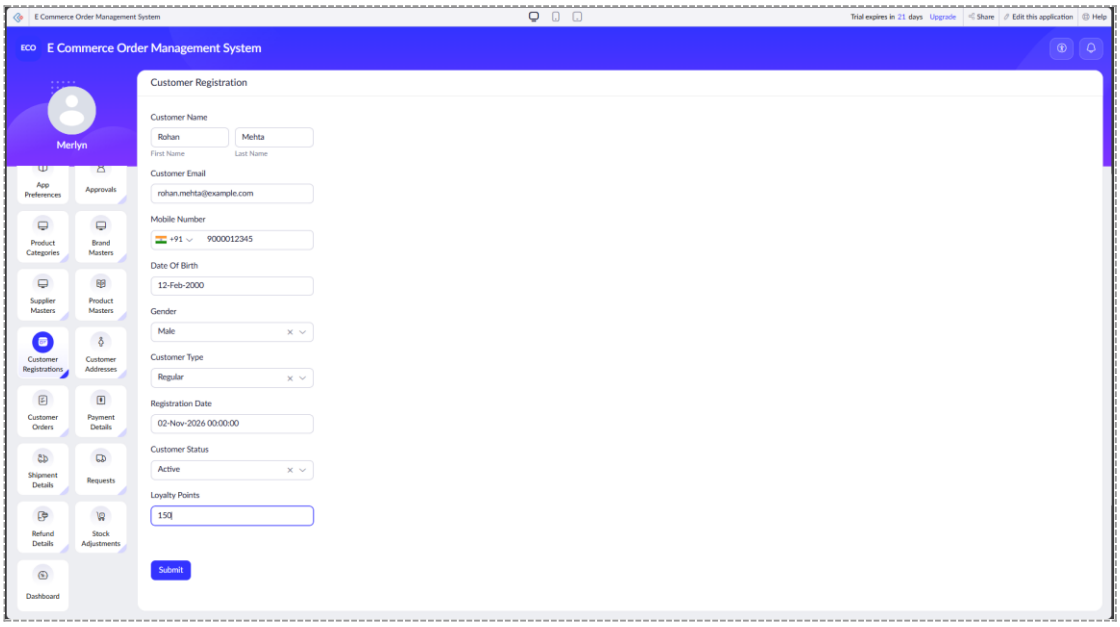


Figure 8.1: Scenario Step 1 – Register Customer and Add an Order

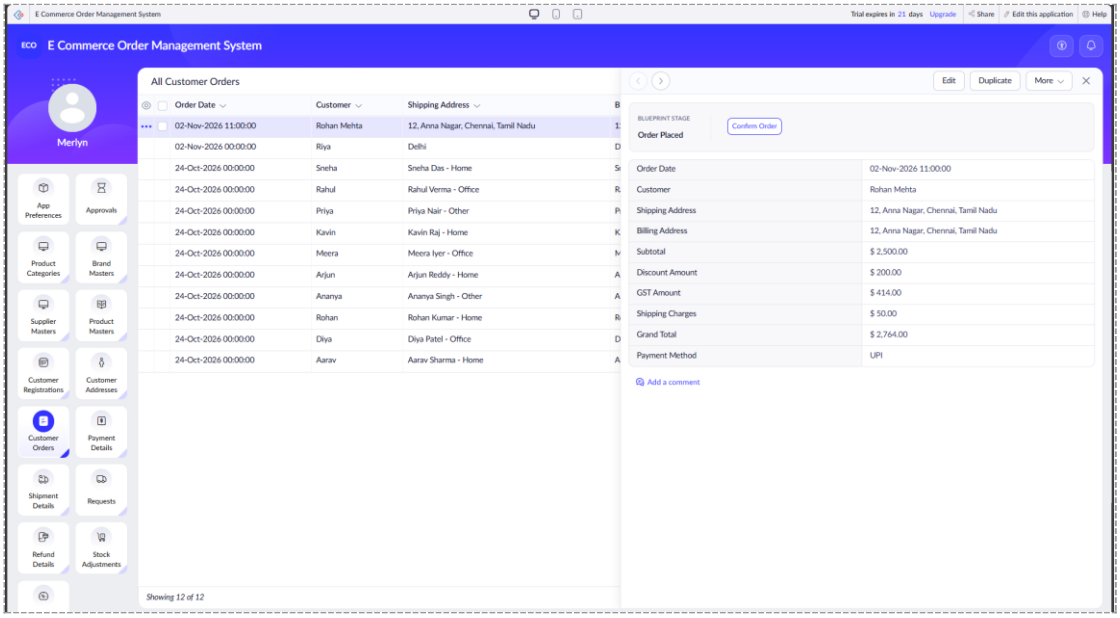


Figure 8.2: Scenario Step 2 – Grand Total Calculated and Blueprint Transition

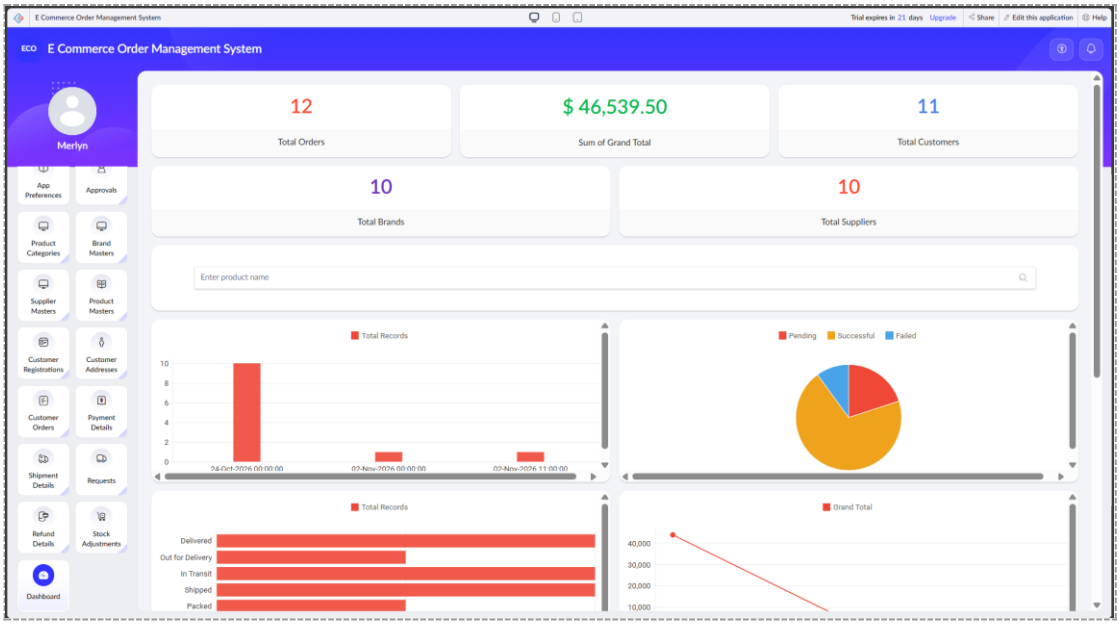


Figure 8.3: Scenario Step 3 – Dashboard Updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Evidence
Connected forms for all record types	12 forms built	Tables 4.2–4.13
Reports and dashboard	28 reports and 1 dashboard page	Table 5.1, Table 5.2
Grand total by Deluge	Workflow configured	DS-01
Order lifecycle Blueprint	5 stages and 4 transitions	Figure 5.8
Approvals and schedules	2 approvals and 3 schedules enabled	Figures 5.6, 5.7
CRM integration	Connection active; record creation unverified	Figure 5.9
Role-based access	5 roles, 7 profiles	Section 5.8

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing System	Proposed Application
Data entry	Several spreadsheets	Connected forms with defaults
Order total	Manual	Calculated by Deluge
Order status	Informal	Blueprint stages
Refund and stock control	No approval	Approval workflows
Reporting	Manual compilation	28 reports and dashboard
Access	Shared file	Role-based profiles

No measured reduction in processing time is claimed at this stage.

8.4 Limitations of the Low-Code Approach

- Platform limits on trial accounts and feature tiers constrain what can be deployed.
- Currency fields are USD while GST rates are Indian; a real deployment should use INR.
- Several cross-references are text rather than lookups, which weakens integrity until they are converted.
- Complex integrations (payment gateway, courier API) need pro-code extension.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of this application was built without code through form, report and page builders and configured workflows. Deluge was needed for the calculation, defaults and the CRM call, which places the project in the low-code part of the spectrum. Pro-code was not required, but would be the next step for external APIs.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The E-Commerce Order Management System brings products, customers, orders, payments, shipments, returns, refunds and stock into one Zoho Creator application. It provides twelve forms, twenty-eight reports, a dashboard, a Deluge total calculation, an order Blueprint, two approval processes, three scheduled reminders, a CRM connection and a role-based security model. The structure and configuration are complete; successful operation of each workflow and the CRM action must still be verified in the live application.

9.2 Future Enhancements

- Convert text references (customer, order, product, brand, supplier) to lookup fields and add an Order Items subform.
- Reduce stock automatically when a shipment is created and on approved stock adjustments.
- Switch currency to INR and compute GST from the product GST rate.
- Integrate a payment gateway and a courier tracking API.
- Enable the customer portal with own-record filters and order tracking.
- Add Zia-based analysis of return descriptions and reorder prediction.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling, form and report design, Deluge scripting, workflow and approval design, Blueprint modelling, integration and access control. These skills apply to business-application development, ERP/CRM customisation and process-automation roles.

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[1] Zoho Corporation, “Zoho Creator Help,” 2026. [Online]. Available: https://www.zoho.com/creator/help/

[2] Zoho Corporation, “Deluge Scripting Language Reference,” 2026. [Online]. Available: https://www.zoho.com/deluge/help/

[3] Zoho Corporation, “Blueprint and Approvals in Zoho Creator,” 2026. [Online]. Available: https://www.zoho.com/creator/help/

[4] R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, “Low-Code as Enabler of Digital Transformation in Manufacturing Industry,” Applied Sciences, vol. 10, no. 1, p. 12, 2020.

[5] Gartner, “Magic Quadrant for Enterprise Low-Code Application Platforms,” 2023.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Status is based on the application export. “Verify” marks features to be demonstrated live.

Table A.1: Feature Coverage Checklist

Feature	Used?	Where / Evidence
Application setup, icon, date and time zone	Yes	Section 5.1
Forms with unique and mandatory validation	Yes	Tables 4.2–4.13
Lookup fields	Yes (2)	Customer Address, Stock Adjustment
Auto Number	Yes (3)	Category, Brand, Supplier IDs
Address field with coordinates	Yes	Customer Address
Subforms	No	Future enhancement
List reports	Yes (12)	Table 5.1
Kanban reports	Yes (8)	Table 5.1
Spreadsheet reports	Yes (4)	Table 5.1
Calendar / Timeline / Map reports	Yes (1/2/1)	Table 5.1
Pages / dashboard with charts	Yes	Table 5.2
Deluge in form workflows	Yes	Table 5.4
Schedule workflows	Yes (3)	Figure 5.6
Approval workflows	Yes (2)	Figure 5.7
Blueprint	Yes	Figure 5.8
Zoho CRM integration	Configured	Figure 5.9; verify record creation
Roles and permission profiles	Yes	Section 5.8
Customer portal	Profile only	Verify live
Zia / AI features	Yes	Section 5.9
Pro-code extension	No	Section 5.10
Batch workflows / Payments	No	Not required

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

Course Outcome (CO)	Chapters / Sections
CO1 – Set up Zoho Creator and understand low-code concepts	Chapters 1, 2; Section 5.1
CO2 – Model data, build forms, reports and pages; write Deluge	Chapters 4, 5 (5.2–5.5)
CO3 – Manage users, roles, AI and analytics	Sections 4.6, 5.8, 5.9
CO4 – Automate with workflows, approvals, Blueprint and integrations	Sections 5.6, 5.7
CO5 – Deploy, secure and extend with pro-code	Chapter 7; Section 5.10

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Calculate Grand Total (Customer Order, on add or edit, on success)

```
Grand_Total = ifnull(input.subtotal,0) -
ifnull(input.discount_amount,0)
             + ifnull(input.gst_amount,0) +
ifnull(input.shipping_charges,0);
```

C.2 Set default dates (on add, on load)

```
// Customer_Registration (registration_date), Customer_Order
(order_date),
// Payment_Details (payment_date), Shipment_Details (shipment_date),
// Return_Request (return_request_date), Stock_Adjustment
(adjustment_date)
if(input.registration_date == null)
{
    input.registration_date = zoho.currentdate;
}
```

C.3 Set default status (on add, on validate)

```
// Customer_Registration
if(input.customer_status == null)
{
    input.customer_status = "Active";
}
// Brand_Master (Brand_Status), Product_Category (Category_Status),
// Supplier_Master (Supplier_Status) use the same pattern.
```

C.4 Sync Customer to Zoho CRM (Customer Registration, on add, on success)

```
ZohoCRM.v2.createContact(data:{"Email":"customer_email",
    "Last_Name":"customer_name.last_name",
    "Phone":"mobile_number"});
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application at *https://merlynsosasaju2024cser72.zohocreatorportal.in* and sign in with the credentials for your role. The Dashboard shows summary indicators.

D.2 Instructions for Order Manager

1. Open Customer Registration and add the customer; open Customer Address to add addresses.
2. Open Customer Order, enter amounts and submit; the grand total is calculated.
3. Open the order and use the Blueprint buttons Confirm Order, then later Mark Delivered.
4. Use All Customer Orders to review orders.

D.3 Instructions for Warehouse Staff

1. Maintain products in Product Master.
2. Record shipments in Shipment Details with courier and tracking number.
3. Add a Stock Adjustment; wait for approval; check the e-mail.

D.4 Instructions for Finance and Support

1. Support raises a Return Request for an order.
2. Finance adds Refund Details; refunds with status Pending are sent for approval.
3. Check the decision e-mail and update the refund status.

APPENDIX E – EVALUATION RUBRIC

The rubric below is retained from the course template. Marks are awarded by the examiners.

Table E.1: Evaluation Rubric for the Individual Project

Criterion	Max Marks	Awarded
Problem definition and requirements	10	
Data model and forms	15	
Reports and dashboards	10	
Deluge scripting	15	
Workflows, approvals, Blueprint	15	
Integration and security	10	
Testing and documentation	10	
Demonstration and viva	15	

APPENDIX F – VIVA-VOCE RECORD SHEET

Table F.1: Viva-Voce Questions and Responses

S.No.	Question	Response Summary	Marks
1			
2			
3			
4			
5			

Table F.2: Marks Summary

Component	Marks
Report and application	
Demonstration	
Viva-voce	
Total	